

RC-GRPO: Reconstruction-Calibrated Temporal Credit Assignment for Tokenized Video World Models

Anonymous submission

Abstract

Introduction

Video world models predict future visual observations conditioned on past frames and actions, providing sampleable environment models for robotics, interactive simulation, and model-based reinforcement learning (Ha and Schmidhuber 2018; Hafner et al. 2019, 2024; He et al. 2026). A common implementation tokenizes visual observations and casts action-conditioned video prediction as sequence modeling: frames are encoded into discrete visual tokens, an autoregressive model predicts future tokens, and a decoder maps generated tokens back to images (Yan et al. 2021; Yu et al. 2023; Wu et al. 2024; Mentzer et al. 2024). Recent verifiable reward frameworks further suggest that these models can be improved after pre-training by sampling a group of candidate futures, scoring decoded predictions against ground-truth frames, and applying GRPO to reinforce candidates with higher within-group rewards (Wu et al. 2025; Shao et al. 2024). This pipeline is attractive because it optimizes against measurable prediction errors rather than a learned critic.

This paper studies a narrower but important question: when the world model is tokenized, is the decoded full-reference reward the right signal for GRPO, especially in multi-step rollouts? We find two coupled failure modes. First, post-decode verifiers compare candidates with raw ground truth even though the ground truth is not exactly reachable through the tokenizer-decoder pair. Second, multi-step GRPO often collapses a rollout into one sequence-level advantage, even though video prediction is autoregressive and frame errors propagate across horizons. These failures are easy to miss because the training loop remains verifiable: every candidate is still compared with a known target.

The first failure mode is verifier-side. A lossy visual tokenizer imposes an irreducible encode-decode residual on the ground-truth frame. We call this residual the reconstruction floor. It is not caused by any sampled candidate, but post-decode metrics such as MSE, SSIM, and LPIPS still

include it when scoring decoded predictions (Wang et al. 2004; Zhang et al. 2018). For ordinary evaluation, this is a reconstruction-quality limit of the tokenizer (van den Oord, Vinyals, and Kavukcuoglu 2017; Esser, Rombach, and Ommer 2021; Blau and Michaeli 2018). For GRPO, it is a ranking problem: the update depends on relative rewards within a sample group, so reconstruction-induced noise can change which candidate receives positive advantage.

The second failure mode is temporal. In a multi-step video rollout, early generated frames become the context for later predictions. A late error can therefore reflect earlier choices, later choices, or their interaction. Sequence-level GRPO ignores this structure by assigning the same normalized advantage to all generated tokens in the rollout. Fine-grained credit assignment has been explored for GRPO-style training in language reasoning and generative trajectories (Parthasarathi et al. 2025; Guo et al. 2025; Kim et al. 2026; Savani et al. 2026; Yu et al. 2025), but tokenized video has a different natural unit of credit: a future frame is represented by a block of visual tokens and evaluated by a frame-level verifier.

We propose *RC-GRPO*, a reconstruction-calibrated temporal credit assignment method for tokenized video world models. RC-GRPO keeps the sample-group, verifiable-reward, and group-relative update structure intact, but changes the target and the credit unit. For verifier calibration, it replaces the raw ground-truth frame with the tokenizer-reachable target $\tilde{s}_t = D(E(s_t))$, so decoded predictions are compared in the same reachable image space. For temporal credit assignment, it computes frame-level rewards in multi-step rollouts and forms temporal-return advantages for each future-frame token block. Early tokens can then receive credit through later frame quality, while horizon-specific prediction errors are no longer mixed into a single scalar advantage.

Our contributions are threefold. First, we identify a reconstruction-floor failure mode in tokenized video GRPO and verify that the resulting pairwise rank flips follow the predicted $\arccos(\rho)/\pi$ relationship across several reward spaces. Second, we introduce reconstruction-calibrated verifiers as the reward substrate for both single-step and multi-step training. Third, we introduce temporal-return GRPO for multi-step video rollouts and show, under a fixed held-out protocol on RT-1 prediction, that it improves full-reference fidelity over sequence-level GRPO, with gains increasing toward later horizons. We also report negative auxiliary results

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for generic GRPO replacements, which supports the conclusion that the effective intervention point is video-structured calibration and temporal credit rather than a generic optimizer swap.

Related Work

Tokenized video world models. World models learn predictive environment dynamics that can support planning, control, and simulation from visual observations (Ha and Schmidhuber 2018; Hafner et al. 2019, 2021, 2024). Recent video generative models extend this idea by modeling future visual observations directly, often through tokenized sequence modeling. VideoGPT, MAGVIT, iVideoGPT, Genie, and pre-trained video simulators show that discrete visual tokens and autoregressive or masked sequence models can support scalable video prediction and interactive generation (Yan et al. 2021; Yu et al. 2023; Wu et al. 2024; Bruce et al. 2024; He et al. 2026). These works mainly study how to build or pre-train a sampleable video world model. Our focus is different: given a tokenized video world model and a verifiable post-training loop, we study whether the decoded verifier target and the sequence-level GRPO update match the structure of tokenized multi-step video prediction.

Verifiable reward post-training. Verifiable reward training samples multiple candidate outputs for the same context, scores them with an externally computable reward, and applies group-relative policy updates without a learned value function. RLVR-World instantiates this idea for world models, while GRPO and related RLVR-style methods have been widely explored in language reasoning and post-training (Wu et al. 2025; Shao et al. 2024; DeepSeek-AI et al. 2025). Recent variants analyze length bias, clipping behavior, large-scale optimization, classification-style rank labels, and adaptive policy objectives (Liu et al. 2025; Yu et al. 2025; Zheng et al. 2025; Zhai et al. 2026; Gao et al. 2025). We do not treat these optimizer variants as direct solutions to tokenized video post-training. In our setting, the verifier itself can corrupt within-group rankings because decoded candidates are compared with targets that are not exactly reachable through the tokenizer-decoder pair. This shifts the relevant intervention from a generic optimizer swap to video-structured verifier calibration.

Visual tokenizers and evaluation metrics. Discrete visual representation learning compresses images or videos into token sequences that can be modeled by transformers, but lossy tokenization introduces an encode-decode reconstruction limit (van den Oord, Vinyals, and Kavukcuoglu 2017; Esser, Rombach, and Ommer 2021; Mentzer et al. 2024). This limit is closely related to the broader perception-distortion tradeoff in image reconstruction (Blau and Michaeli 2018). Full-reference metrics such as SSIM and LPIPS are commonly used to evaluate per-sample fidelity, whereas FID, KID, FVD, DINO-based distances, precision/recall, and density/coverage evaluate distribution-level fidelity and diversity (Wang et al. 2004; Zhang et al. 2018; Heusel et al. 2017; Bińkowski et al. 2018; Unterthiner et al. 2018; Oquab et al. 2024; Kynkäänniemi et al. 2019; Naeem

et al. 2020). These metrics are usually used for reporting generation quality. We instead ask what happens when a post-decode full-reference metric becomes the reward consumed by GRPO: distribution-level metrics do not directly provide a per-candidate verifier, while post-decode per-sample metrics inherit the tokenizer reconstruction floor and can alter candidate rankings.

Fine-grained credit assignment. Sequence-level advantages can be too coarse when a generation trajectory contains multiple reasoning steps, semantic segments, or generation stages. Recent GRPO-style work addresses this issue through eligibility-trace credit assignment, segment-level policy optimization, verifiable segment decomposition, stepwise credit for flow-matching generation, and token-level loss shaping in large-scale RL systems (Parthasarathi et al. 2025; Guo et al. 2025; Kim et al. 2026; Savani et al. 2026; Yu et al. 2025). RC-GRPO adapts this principle to tokenized video rather than reusing token-level or denoising-step credit directly. In a video rollout, the natural credit unit is a future-frame block of visual tokens, and each block is scored by a frame-level full-reference verifier. This motivates temporal-return advantages over frame blocks after reconstruction-calibrated verifier targets, rather than a task-agnostic GRPO objective.

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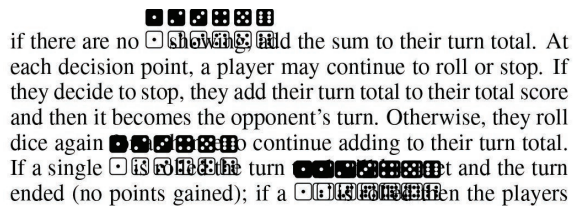
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if there are no `\showbox` and the sum to their turn total. At each decision point, a player may continue to roll or stop. If they decide to stop, they add their turn total to their total score and then it becomes the opponent's turn. Otherwise, they roll dice again `\showbox` to continue adding to their turn total. If a single `\showbox` turn `\showbox` and the turn ended (no points gained); if a `\showbox` then the players

Figure 1: Using the trim and clip commands produces fragile layers that can result in disasters (like this one from an actual paper) when the color space is corrected or the PDF combined with others for the final proceedings. Crop your figures properly in a graphics program – not in $\text{\LaTeX}$.

If you normally create your figures using `pgfplots`, note that `pgfplots` itself must not be used in your paper source (see Table 2). Pre-generate your figures externally, export them as PDF files, and import them with `\includegraphics`. Bounding and trim boxes created by `pgfplots` are fragile and tend to create issues.

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Figure Captions. The illustration number and caption must appear *under* the illustration. Labels and other text with the actual illustration must be at least nine-point type. However, the font and size of figure captions must be 10 point roman. Do not make them smaller, bold, or italic. (Individual words may be italicized if the context requires differentiation.)

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Tables that do not fit in a single column must be placed across double columns. If your table won't fit within the margins even when spanning both columns and using the above techniques, you must split it in two separate tables.

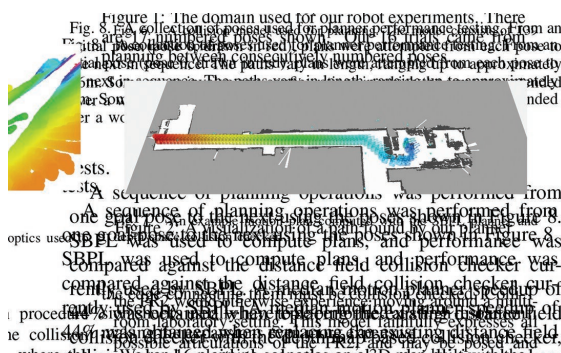


Figure 1: The domain used for our robot experiments. There are 8 planning operations performed from an initial pose to a goal pose. The path is shown in yellow. The path is labeled with numbers 1 through 8, indicating a sequence of planning operations. The path is shown in a 3D perspective view.

Grid Search Domain

B. View Selection

In these experiments, we ran our planner on a much larger set of the experiments described here, but with manual selection of views from which depth maps were captured. The only difference between these experiments and the previous ones is that we used a manual selection of views from which depth maps were captured. This only affects the performance of our planner. We ran these trials on a machine with a 2.4 GHz processor and 4 GB of RAM. The results are shown in Table 1. The results show that the manual selection of views from which depth maps were captured significantly improves the performance of our planner. The results also show that the manual selection of views from which depth maps were captured significantly improves the performance of our planner. The results also show that the manual selection of views from which depth maps were captured significantly improves the performance of our planner.

Figure 2: Adjusting the bounding box instead of actually removing the unwanted data resulted multiple layers in this paper. It also needlessly increased the PDF size. In this case, the size of the unwanted layer doubled the paper’s size, and produced the following surprising results in final production. Crop your figures properly in a graphics program. Don’t just alter the bounding box.

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LaTeX Overflow. LaTeX users please beware: LaTeX will sometimes put portions of the figure or table or an equation in the margin. If this happens, you need to make the figure or table span both columns. If absolutely necessary, you may reduce the figure, or reformat the equation, or reconfigure the table. **Check your log file!** You must fix any overflow into the margin (that means no overfull boxes in LaTeX). **Nothing is permitted to intrude into the margin or gutter.**

Using Color. Use of color should be limited to figures and minor text details (it is ok to colorize one word, but not an entire sentence). It must be WCAG 2.0 compliant. (That is, the contrast ratio must be greater than 4.5:1 no matter the font size.) It may never be used for any portion of the text of your paper. The archival version of your paper may be printed in grayscale. The web version should be readable by persons with disabilities. Consequently, because conversion to grayscale can cause undesirable effects (red changes to black, yellow can disappear, and so forth), we strongly suggest you test the figures in your document when converted to grayscale. Also, you should be mindful of readers who may happen to have trouble distinguishing colors. Hence, your paper must be decipherable without using color for distinc-

tion.

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Fonts in Your Illustrations. You must embed all fonts in your graphics before including them in your LaTeX document.

Algorithms. Algorithms and/or programs are a special kind of figures. Like all illustrations, they should appear floated to the top (preferably) or bottom of the page. However, their caption should appear in the header, left-justified and enclosed between horizontal lines, as shown in Algorithm 1. The algorithm body should be terminated with another horizontal line. It is up to the authors to decide whether to show line numbers or not, how to format comments, etc.

In LaTeX algorithms may be typeset using the algorithm and algorithmic packages, but you can also use one of the many other packages for the task.

Listings. Listings are much like algorithms and programs. They should also appear floated to the top (preferably) or

Algorithm 1: Example algorithm

Input: Your algorithm's input

Parameter: Optional list of parameters

Output: Your algorithm's output

```
1: Let  $t = 0$ .
2: while condition do
3:   Do some action.
4:   if conditional then
5:     Perform task A.
6:   else
7:     Perform task B.
8:   end if
9: end while
10: return solution
```

Listing 1: Example listing `quicksort.hs`

```
1 quicksort :: Ord a => [a] -> [a]
2 quicksort [] = []
3 quicksort (p:xs) = (quicksort lesser) ++
4   [p] ++ (quicksort greater)
5   where
6     lesser = filter (< p) xs
7     greater = filter (>= p) xs
```

bottom of the page. Listing captions should appear in the header, left-justified and enclosed between horizontal lines as shown in Listing 1. Terminate the body with another horizontal line and avoid any background color. Line numbers, if included, must appear within the text column.

References

The AAAI style includes a set of definitions for use in formatting references with BibTeX. These definitions make the bibliography style fairly close to the ones specified in the Reference Examples appendix below. To use these definitions, you also need the BibTeX style file “`aaai2027.bst`,” available in the AAAI Author Kit on the AAAI web site. Then, at the end of your paper but before `\enddocument`, you need to put the following lines:

```
\bibliography{bibfile1,bibfile2,...}
```

Please note that the `aaai2027.sty` class already sets the `\bibliographystyle` for you, so you do not have to place any `\bibliographystyle` command in the document yourselves. The `aaai2027.sty` file is incompatible with the `hyperref` and `navigator` packages. If you use either, your references will be garbled and your paper will be returned to you.

References may be the same size as surrounding text. However, in this section (only), you may reduce the size to `\small` (9pt) if your paper exceeds the allowable number of pages. Making it any smaller than 9 point with 10 point linespacing, however, is not allowed.

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The following commands are available for your use in citing references:

`\cite`: Cites the given reference(s) with a full citation. This appears as “(Author Year)” for one reference, or “(Author Year; Author Year)” for multiple references.

`\shortcite`: Cites the given reference(s) with just the year. This appears as “(Year)” for one reference, or “(Year; Year)” for multiple references.

`\citeauthor`: Cites the given reference(s) with just the author name(s) and no parentheses.

`\citeyear`: Cites the given reference(s) with just the date(s) and no parentheses.

You may also use any of the *natbib* citation commands.

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Please check all the pages of your PDF file. The most commonly forgotten element is the acknowledgments — especially the correct grant number. Authors also commonly forget to add the metadata to the source, use the wrong reference style file, or don't follow the capitalization rules or comma placement for their author-title information properly. A final common problem is text (especially equations) that runs into the margin. You will need to fix these common errors before submitting your file.

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Formatted bibliographies should look like the following examples. You should use BibTeX to generate the references. Missing fields are unacceptable when compiling references, and usually indicate that you are using the wrong type of entry (BibTeX class).

Book with multiple authors Use the `@book` class.

Engelmore, R.; and Morgan, A., eds. 1986. *Blackboard Systems*. Reading, Mass.: Addison-Wesley.

Journal and magazine articles Use the `@article` class.

Robinson, A. L. 1980. New Ways to Make Microcircuits Smaller. *Science*, 208(4447): 1019–1022.

Hasling, D. W.; Clancey, W. J.; and Rennels, G. 1984. Strategic explanations for a diagnostic consultation system. *International Journal of Man-Machine Studies*, 20(1): 3–19.

Proceedings paper published by a society, press or publisher Use the `@inproceedings` class. You may abbreviate the *booktitle* field, but make sure that the conference edition is clear.

Clancey, W. J. 1984. Classification Problem Solving. In *Proceedings of the Fourth National Conference on Artificial Intelligence*, 45–54. Menlo Park, Calif.: AAAI Press.

Clancey, W. J. 1983. Communication, Simulation, and Intelligent Agents: Implications of Personal Intelligent Machines for Medical Education. In *Proceedings of the Eighth International Joint Conference on Artificial Intelligence (IJCAI-83)*, 556–560. Menlo Park, Calif: IJCAI Organization.

University technical report Use the `@techreport` class.

Rice, J. 1986. Polygon: A System for Parallel Problem Solving. Technical Report KSL-86-19, Dept. of Computer Science, Stanford Univ.

Dissertation or thesis Use the `@phdthesis` class.

Clancey, W. J. 1979. *Transfer of Rule-Based Expertise through a Tutorial Dialogue*. Ph.D. diss., Dept. of Computer Science, Stanford Univ., Stanford, Calif.

Forthcoming publication Use the `@misc` class with a `note="Forthcoming"` annotation.

```
@misc(key,
[...],
note="Forthcoming",
)
```

Clancey, W. J. 2021. The Engineering of Qualitative Models. Forthcoming.

ArXiv paper Fetch the BibTeX entry from the “Export Bibtex Citation” link in the arXiv website. Notice it uses the `@misc` class instead of the `@article` one, and that it includes the `eprint` and `archivePrefix` keys.

```
@misc(key,
[...],
eprint="xxxx.yyyy",
archivePrefix="arXiv",
)
```

Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, L.; and Polosukhin, I. 2023. Attention Is All You Need. arXiv:1706.03762.

Website or online resource Use the `@misc` class. Add the url in the `howpublished` field and the date of access in the `note` field:

```
@misc(key,
[...],
howpublished="\url{http://...}",
note="Accessed: YYYY-mm-dd",
)
```

NASA. 2015. Pluto: The ‘Other’ Red Planet. <https://www.nasa.gov/nh/pluto-the-other-red-planet>. Accessed: 2018-12-06.

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Thank you for reading these instructions carefully. We look forward to receiving your electronic files!

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